

Conduct Guardian

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CONTENTS

Conduct Guardian	
The problem	
What it does	
Architecture	
Measured results	
Where the rules come from	
Honest limitations	
Reproduce any number above	

Conduct Guardian

Conduct-compliance screening for Malaysian debt collection, built against the Consumer Credit Act 2025 (Act 873).

The problem

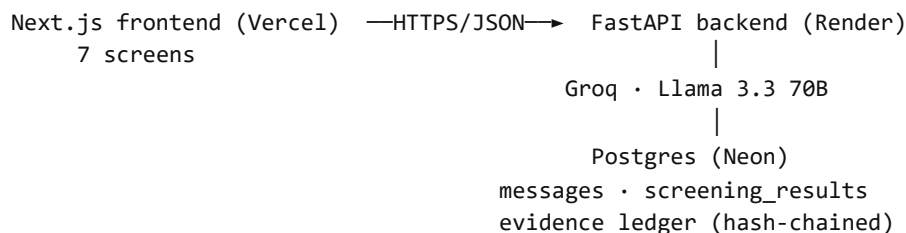
Act 873 came into force **1 March 2026**. Since **1 June 2026**, collection agencies must register with the Consumer Credit Commission and *prove* fair conduct. Non-compliance carries fines up to **RM5 million and/or 5 years**, plus a mandatory compliance audit under sections 49 and 66.

Conduct happens across WhatsApp, SMS, phone and email. No agency can currently see across all of it — so none of them can prove anything.

What it does

Screens every collector→customer message against a conduct rule pack, flags violations with the exact offending phrase and a compliant rewrite, merges all channels into one per-account timeline to catch cross-channel harassment, writes every verdict to a tamper-evident hash-chained evidence ledger, turns repeat violations into targeted coaching, and routes borrowers showing genuine hardship to human review instead of continued escalation.

Architecture



One real pipeline. `POST /screen` is the only write path: message → LLM screening → stored with a chained ledger entry. The seed script drives that *same public endpoint over HTTP* rather than calling the code directly, so every verdict in the database is a genuine model output. Every dashboard number is a SQL aggregate over that one dataset — nothing is faked per screen.

Measured results

All figures below were produced by scripts in the repo and are reproducible.

Metric	Result	Notes
Agreement with human labels	96.6%	llama-3.3-70b, 59 collector messages
Golden-set precision / recall / F1	91.7% / 100% / 95.7%	llama-3.1-8b, 20 curated cases
False negatives	0	No real violation missed — the right way to err
False positives	1	A co-signing guarantor case; arguably a legitimate party
Prompt-injection battery	7 / 7 held	Incl. forged delimiter, Malay-language override, invented rule ID
Invalid rule IDs returned	0	Post-validation rejects anything not in the pack
Live screening latency	947 ms	llama-3.3-70b, end to end
Seeded corpus	75 / 75 , 0 failures	Every verdict from the real endpoint
Evidence ledger	valid , 0 broken rows	Rehashed from stored payloads
Automated tests	27 passing	Incl. direct- UPDATE tamper detection

Where the rules come from

The Commission's **Conduct Standards v1.0 (5 June 2026)** exists but is classified **Restricted (Terhad)** and is not publicly retrievable. We did not use it.

The 8-rule pack instead derives from **Act 873 s.85(1)(g)** (fair debt collection practices) and **Bank Negara Malaysia's published Fair Debt Collection Practices** guidance. **Every rule cites its source.**

The pack is data, not prompt text — a registered agency swaps it for the Restricted standards under its registration and the engine is unchanged.

One consequence worth noting: the contact-frequency rule uses BNM's published **maximum 3 contacts per week**, computed deterministically in SQL rather than judged by the model. A separate same-hour burst check exists but is labelled everywhere as an internal heuristic, never as a regulatory limit.

Honest limitations

Stated up front rather than discovered under questioning.

- **The hash chain proves internal consistency, not third-party non-repudiation.** Anyone with write access to the whole table could recompute it end to end. Production anchors the head hash where the agency cannot rewrite it — notarisation, a signed periodic head, or WORM storage. That is a deployment change, not a redesign.
- **The rate limiter is in-process** — it resets on restart and does not coordinate across replicas. Correct for a single instance; Redis if it scales.
- **Call transcripts are typed text “as if transcribed”.** No audio ingestion.
- **No production auth, no multi-tenancy.** Both explicitly out of scope.
- **All data is synthetic.** No real borrower, collector, or agency data is used anywhere, ever.
- The bulk model (llama-3.1-8b) is materially weaker than llama-3.3-70b — 85.5% vs 96.6% agreement. It is used for seeding only; live screening runs on 70B.

Reproduce any number above

```
cd backend
./venv/Scripts/python.exe -m pytest -q           # 27 tests
./venv/Scripts/python.exe -m scripts.eval_golden_set # precision / recall / F1
./venv/Scripts/python.exe -m scripts.probe_injection # 7 injection attacks
curl -X POST $API/ledger/verify                   # rehash the whole chain
```